

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2226

Crawfordsville Power Plant 138 kV
July 19, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2226. The Interconnection Customer proposes a 19.6 MW battery energy storage facility interconnecting at the Crawfordsville Power Plant 138 kV substation in the New Jersey area. The request was received on November 09, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 530 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,116,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mt Poso - W Orange 138kV	Hilltop Alte - Blanchard 138kV	142.9	230	8.1
Whippany - Orange 138kV	Blanchard - Hilltop Alte 138kV	130.3	328	8.1
Sayreville - Edison 138kV	Allston 345-138 kV XFMR	117.8	274	12.6
Mt Poso Substation - Whippany 138kV	Hilltop Alte - Blanchard 138kV	112.5	346	8.1
Allegany Cogen - Sayreville 138kV	Allston 345 kV/138 kV Transformer	112.4	251	12.6
Morfee - Pseg Linden Generating Station 161kV line	Pseg Linden Generating Station - Morfee 161kV	110.4	452	6.4
Allegany Cogen - Allston 138kV	Allston 345/138kV XFMR	109.2	258	12.6
Aldene - Beaver Dam 3Rd St 138kV Line	Magnolia 1708 - White Rock-Nike 138kV L	107.4	255	5.8
Orange 345-138 kV Transformer Bank	Laurel Ave - Orange 345kV Line	104.1	598	8.2
Locust Grove 345/138kV Auto	W Orange - Magnolia 1708 138kV	103.2	506	13.7
Morfee - Pseg Linden Generating Sta 161kV	Pseg Linden Generating Station - Morfee 161kV Circuit 1	101.9	490	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mt Poso - W Orange 138kV	Rebuild line with higher-capacity conductor	\$2,965,000	15% (\$436,000)
Whippany - Orange Substation 138kV	Rebuild line with higher-capacity conductor	\$2,902,000	15% (\$427,000)
Sayreville - Edison 138kV	Rebuild line with higher-capacity conductor	\$2,842,000	2% (\$61,000)
Whippany - Mt Poso 138kV	Rebuild line with higher-capacity conductor	\$3,847,000	15% (\$566,000)
Allegany Cogen - Sayreville 138kV	Rebuild line with higher-capacity conductor	\$1,886,000	2% (\$40,000)
Morfee - Pseg Linden Generating Station 161kV line	Rebuild line with higher-capacity conductor	\$1,868,000	100% (\$1,868,000)
Allegany Cogen - Allston 138kV	Reconductor line	\$429,000	2% (\$9,000)
Aldene - Beaver Dam 3Rd St 138kV	Reconductor line	\$592,000	100% (\$592,000)
Orange 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$15,006,000	2% (\$311,000)
Locust Grove 345/138kV XFMR	Replace transformer with higher-rated unit	\$13,802,000	16% (\$2,185,000)
Morfee - Pseg Linden Generating Sta 161kV	Reconductor line	\$621,000	100% (\$621,000)
	Total allocated to project		\$7,116,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.